ClaimFlow: Tracing the Evolution of Scientific Claims in NLP

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Abstract

Scientific papers advance *claims* that later work supports, extends, or sometimes refutes. Yet existing methods for citation and claim analysis capture only fragments of this dialogue. In this work, we make these interactions explicit at the level of scientific claims. We introduce ClaimFlow, a claim-centric view of the NLP literature, built from 1,617 ACL Anthology papers (1979–2025) that are manually annotated with 5,689 claims and 4,871 cross-paper claim relations, indicating whether a citing paper supports, extends, qualifies, refutes, or references a cited claim as background. Building on ClaimFlow, we define a new task – *Claim Relation Classification* – which requires models to infer the scientific stance toward a cited claim from the text and citation context. Evaluating neural models and large language models on this task, we report baseline performance of 0.81 macro-F1, suggesting that the task is tractable while leaving room for improvement. We then scale this framework to $\sim 13k$ NLP papers to study claim evolution across decades of NLP research. We show that 63.5% of claims are never reused; only 11.1% are ever challenged. Widely propagated claims are more often *reshaped* through qualification and extension than supported or refuted. Overall, ClaimFlow offers a lens for examining how ideas shift and mature within NLP. [1]

1 Introduction

Scientific research is often described as a steady accumulation of knowledge, but what actually accumulates are *claims*: assertions about how the world works, why a method succeeds, or under what conditions an idea fails. These claims are the basic currency of scientific inquiry (Lakatos, 1976; Popper, 2002). A new paper often includes claims that reinforce, extend (qualify), or even challenge (overturn) past claims, yet these interactions remain poorly understood.

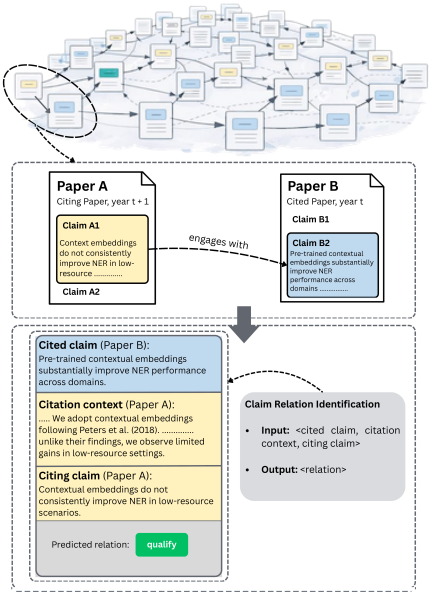


Figure 1: Claims form a directed graph across papers: edges link citing and cited claims, and their relations are predicted from claims and citation context.

NLP is a particularly interesting field to study this process. It spans computer science, linguistics, and social science (Manning and Schutze, 1999; Jurafsky and Martin, 2026); the field has repeatedly refined its central questions (Pramanick et al., 2023). These shifts are reflected in the kinds of scientific claims researchers make: from early symbolic accounts of parsing (Winograd, 1972) to statistical alignment models (Brown et al., 1993) to deep contextual representations (Devlin et al., 2019), and more recently to claims about scaling laws and reasoning abilities in LLMs (Wei et al., 2022). Yet we still lack systematic tools for organizing claims into a coherent record of how the field has developed.

Understanding the scientific character of NLP requires moving beyond papers as atomic units to-

[1] Code and data available at https://github.com/UKPLab/emnlp2026-claim-flow

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ward the claims they advance and how later work engages with them. Each research paper introduces one or more central claims – about model behavior, properties of data and language, or assumptions about human language use – and later citations implicitly position new work relative to these claims. Such engagement may support a claim, extend it to new settings, qualify its scope, or actively refute it.

In this paper, we make claim-level scientific interaction explicit. We introduce `ClaimFlow`, a claim-centric view of the NLP literature constructed from 1,617 ACL Anthology papers from 1979 to 2025 (Bird et al., 2008), manually annotated with 5,689 scientific claims and 4,871 cross-paper claim-level relations. Relations are labeled as `background`, `support`, `extension`, `qualification`, or `refutation`. Building on `ClaimFlow`, we define *Claim Relation Classification*, a task that takes as input (i) a *cited claim* from an earlier paper, (ii) a *citing claim* advanced by a later paper, and (iii) the *citation context* – the sentence containing the citation marker together with its immediately preceding and following sentences – to determine whether later work adopts, extends, or challenges a prior claim. We build on existing claim identification methods (Pramanick et al., 2025) to focus on modeling how claims in later papers engage with prior work. Figure 1 illustrates the claim graph and the Claim Relation Classification task.

To study whether current models can recover claim-level relations, we develop baseline systems for *Claim Relation Classification*. Models achieve reasonable overall performance (e.g., macro-F1 0.81), but struggle with subtle distinctions such as extension versus qualification. We apply the best-performing model to recover claim relations across 13,358 ACL Anthology papers spanning 1979–2025, enabling longitudinal analysis of how scientific ideas evolve within NLP research. Our analyses reveal that challenges are rare and typically short-lived, while widely propagated claims are more often refined through qualification and extension than directly confirmed or rejected.

Contributions. Our contributions are threefold: (1) we introduce `ClaimFlow`, an expert-annotated dataset of 1,617 NLP papers containing 5,689 scientific claims and labeled relations between claims across papers, enabling fine-grained analysis of how claims are reused, extended, or challenged (§3); (2) we define *Claim Relation Classification*, a new task that requires models to infer the stance a citing claim takes toward a cited claim given the citation context (§4); and finally, (3) we leverage this framework to present the first large-scale, claim-level longitudinal analysis of NLP research, examining how scientific claims propagate, are challenged, and evolve across $\sim 13k$ ACL Anthology papers published between 1979 and 2025 (§5).

2 Related Work

Longitudinal Analyses of NLP Research. Several studies have analyzed the evolution of NLP and other scientific fields by examining trends in topics, methods, datasets, or contributions over time (Hall et al., 2008; Xiao et al., 2022; Pramanick et al., 2023, 2025; Abdalla et al., 2023; Wahle et al., 2025). They provide valuable high-level perspectives and often rely on surface-level features such as keywords and metadata. Text mining and deep learning techniques have also been utilized to enable detailed analyses of the interactions among topics and their evolution (Prabhakaran et al., 2016; Tan et al., 2017; Salloum et al., 2017; Hou et al., 2019; Pramanick et al., 2023; Şahinuç et al., 2024). `ClaimFlow` complements this line of work by introducing claims as the units of analysis, enabling finer-grained longitudinal analyses of how scientific ideas emerge, persist, and change.

Citation Analysis. A large body of work analyzes citations as signals of influence or scholarly communication, often using graphs or bibliometric measures (Bornmann and Daniel, 2008). Moving beyond citation counts, prior work has proposed classifying citation intent or function, distinguishing roles such as background, use, or comparison (Teufel et al., 2006; Jurgens et al., 2018; Cohan et al., 2019). These approaches operate at the level of sentences and focus on rhetorical or functional aspects of citations. In contrast, our work models epistemic relations between scientific claims, which captures scientific engagement that cannot be recovered from citation intent labels alone.

Scientific Claim and Argument Mining. Prior work has explored scientific claim extraction, evidence identification, and argument structure modeling in research articles (Al Khatib et al., 2021; Binder et al., 2022; Flashner et al., 2025). Closely related efforts in scientific fact verification and claim checking evaluate whether individual statements are supported by evidence, often within a sin-

Relation	Cited Claim	Citing Claim	Description
SUPPORT	Incorporating a small amount of human-labeled data into distant supervision significantly improves relation extraction performance. (Pershina et al., 2014)	Crowdsourced annotation of training data can significantly improve performance for relation extraction compared to methods based solely on distant supervision. (Liu et al., 2016)	The citing claim provides evidence or results that are consistent with and reinforce the cited claim.
EXTEND	GAN-BERT extends the fine-tuning of BERT-like architectures with unlabeled data in a generative adversarial setting. (Croce et al., 2020)	Extending transformer-based models with Semi-Supervised Generative Adversarial Networks (SS-GAN) improves performance on the Dialect Arabic Identification task. (Yusuf et al., 2022)	The citing claim applies the cited claim to a new setting, task, dataset, or method while preserving its core assertion.
QUALIFY	Neural network models, particularly those based on Long Short-Term Memory (LSTM), can achieve state-of-the-art performance on natural language inference tasks when trained on the SNLI corpus. (Bowman et al., 2015)	The performance of high-performing NLI models is dramatically lower on a subset of examples considered 'hard' compared to the rest of the instances. (Gururangan et al., 2018)	The citing claim restricts, conditions, or refines the scope of the cited claim without rejecting it.
REFUTE	Deeper models can extract more expressive features and improve performance in machine translation tasks. (Bapna et al., 2018)	Increasing the depth of NMT models directly by stacking more blocks does not improve performance and can lead to optimization failure. (Wu et al., 2019)	The citing claim presents evidence or arguments that contradict the cited claim.
BACKGROUND	Byte pair encoding (BPE) is a suitable word segmentation strategy for neural network models. (Sennrich et al., 2016)	BPE-dropout is a simple and effective subword regularization method. (Provilkov et al., 2020)	The cited claim is mentioned for context or positioning, without being evaluated or modified.

Table 1: Illustrative examples of claim–claim relations in `ClaimFlow`, showing how papers engage with prior claims.

gle document or against external sources (Wadden et al., 2020; Saakyan et al., 2021). While these approaches identify claims or argumentative components, they typically treat claims in isolation. Our work instead tracks interactions between claims across papers, explicitly modeling how later work engages with claims introduced in earlier publications.

3 `ClaimFlow`: A Claim-Centric Dataset for Scientific Progress in NLP

To represent scientific progress at the level of individual claims, `ClaimFlow` models how claims are expressed and engaged with across papers.

3.1 Conceptual Framework

3.1.1 What is a Scientific Claim?

In this work, we define a *scientific claim* as a *central, empirically testable assertion about methods, data, or phenomena that a paper advances as part of its contribution*. This definition aligns with established views of scientific progress as the proposal and evaluation of testable assertions (Kuhn and Hacking, 1970; Lakatos, 1976; Swales, 1990; Popper, 2002). We distinguish abstract claims from their textual realizations. A *claim text* refers to a statement that expresses a claim; the same claim may be stated multiple times and in different linguistic forms across sections. In `ClaimFlow`, multiple realizations are normalized into a canonical claim representation, which serves as the unit of annotation and analysis.

3.1.2 What are Claim Relations?

Definition and Scope. A claim relation specifies how a claim from a citing paper engages with a claim from a cited paper. To interpret this interaction, `ClaimFlow` uses the citation context – namely, the sentence containing the citation marker together with its immediately preceding and following sentences – as localized textual evidence linking the two papers.

We treat claim identification as an upstream process and focus on modeling interactions between claims once identified.

Components of Claim–Claim Interaction. As illustrated in Figure 1, determining the relation between two claims requires reasoning over three sources of information: the cited claim, the citing claim, and the local citation context. The cited claim defines the target of engagement, the citing claim expresses the new scientific assertion, and the citation context provides rhetorical evidence for how the prior work is discussed. No single component is sufficient on its own: determining whether an interaction constitutes support, extension, or qualification requires all three jointly.

Textual Grounding. We distinguish *citing and cited claim texts* from the underlying claims they express. Claim relations are defined between claims, but grounded in their textual realizations within papers. A citing claim may occur anywhere in the citing paper and is not necessarily located within the local citation context. When multiple citation contexts refer to the same cited paper, `ClaimFlow` associates the claim pair with the context that most directly expresses the interaction between the claims.

Relation Taxonomy. We define a claim–claim relation taxonomy grounded in prior work on argumentation and scientific discourse (Toulmin,

Relation	Share (%)
SUPPORT	20.9
EXTEND	13.4
QUALIFY	5.4
REFUTE	2.1
BACKGROUND	58.2

Table 2: Percentages of relation types in ClaimFlow.

2003; Lauscher et al., 2018). This taxonomy distinguishes the different ways in which a later claim engages with an earlier claim. ClaimFlow includes five relations: support, extension, qualification, refutation, and background. They capture whether a work reinforces, generalizes, narrows, contradicts, or references a prior claim without substantive evaluation. Table 1 lists definitions and examples.

3.2 Dataset Curation

Data Preparation. To study how scientific claims evolve in NLP, we curate a corpus of papers from the ACL Anthology (Bird et al., 2008). Our corpus consists of 1,617 papers published between 1979 and 2025 from venues that belong to the "ACL Events" category. [2] We sample papers randomly, subject to a minimum of one paper per year.

We retrieve bibliographic metadata from anthology.bib.[3] We parse paper text with Docling (Livathinos et al., 2025) and retrieve citation contexts via the Semantic Scholar API (Lo et al., 2020). We retain only citation links between papers, resulting in a directed citation graph with 2,140 edges. This restriction ensures that all cited claims are available for annotation.

Annotation. Annotation was conducted by two NLP researchers in a two-stage workflow (Cohan et al., 2019; Teufel et al., 2006). **Claim identification.** First, the annotators are provided with the abstract, introduction, and conclusion sections of each paper, and they identify *claim texts* (statements expressing testable assertions about methods, data, or phenomena advanced by the paper). When necessary, annotators make minimal edits (e.g., resolving anaphora) to make the claim texts self-contained. For each claim text, we record its originating sections. **Relation labeling.** In the second phase, annotators examine papers that cite those annotated in the first phase. Annotators inspect the citation context in the citing paper together with the claim texts in both the citing and cited papers, and assign a claim relation label (from Table 1). Ambiguous cases are flagged and resolved through adjudication.

Both annotators label a shared subset of 1,001 papers for agreement evaluation before independently annotating an additional 308 papers each. We refer to the resulting corpus of 1,617 annotated papers as ClaimFlow (see Appendix C for details).

Agreement and Quality. We measure inter-annotator agreement using Krippendorff's α for claim identification, treated as a sentence-level binary labeling task with class imbalance, and Cohen's κ for claim-relation annotation, where annotators assign a single label from a fixed set (Krippendorff, 2004; Artstein and Poesio, 2008). We obtain $\alpha = 0.75$ for claim identification and $\kappa = 0.77$ for claim relation classification, comparable to prior work in scholarly document understanding (Yang and Li, 2018; Hou et al., 2021; Lauscher et al., 2022). The corresponding 95% confidence intervals range from 0.73–0.81 for α and 0.72–0.82 for κ ($p < 0.05$). Relation-wise agreement is reported in Appendix C.7.

3.3 Dataset Statistics

ClaimFlow consists of 1,617 NLP research papers published between 1979 and 2025, annotated with 5,689 scientific claims and 4,871 claim relations inferred from citation contexts. Each paper contributes between 1 and 4 claims, with an average of 3.18 claims per paper.

Claim relations are unevenly distributed across relation types. As shown in Table 2, support and background relations are most common, while extend, qualify, and refute are comparatively rare. The resulting claim graph is substantially sparser than the underlying citation graph, highlighting that citations do not necessarily correspond to substantive claim-level engagement (Nicholson et al., 2021).

4 Automatic Claim Relation Classification

We define *Claim Relation Classification* as the task of predicting how a citing claim engages with a cited claim and benchmark multiple models on ClaimFlow.

[2] Major NLP conferences and workshops, including: ACL, NAACL, EMNLP, EACL, AACL, and Findings.

[3] https://aclanthology.org/anthology.bib.gz

4.1 Task Definition

We formulate the task as a supervised multi-class classification problem. Given a cited claim h_{cited} from paper $\mathcal{B}$, a citing claim h_{citing} from paper $\mathcal{A}$, and a citation context s (defined as the sentence containing the citation marker and its immediately preceding and following sentences), the model predicts the epistemic relation $r \in \{\texttt{background}, \texttt{support}, \texttt{extension}, \texttt{qualification}, \texttt{refutation}\}$ given an input $(h_{\text{cited}}, s, h_{\text{citing}})$.

4.2 Models and Baselines

We evaluate a range of models to assess their ability to predict claim-relations. Our goal is not to exhaustively optimize architectures, but to compare representative modeling paradigms and identify which aspects of the task current models handle well or struggle with.

Encoder-Based Models. We evaluate transformer-based encoder models BERT (Devlin et al., 2019) and RoBERTa (Liu et al., 2019) as supervised baselines for claim relation classification. We additionally evaluate DeBERTa-v3 (He et al., 2021), a stronger encoder architecture with disentangled attention; and to assess the effect of domain-specific pretraining, we evaluate SciBERT (Beltagy et al., 2019). For each instance, we linearize the input triple $(h_{\text{cited}}, s, h_{\text{citing}})$ and feed the resulting sequence into a single encoder. The final hidden representation of the `[CLS]` token is passed to a softmax classifier over the relation label set.

Large Language Models. We also evaluate large language models (LLMs) in a prompt-based setting. For proprietary models, we evaluate GPT-3.5-turbo (Achiam et al., 2023), GPT-4o, and GPT-4.1. For open-source models, we evaluate LLaMA-3-70B-Instruct (Dubey et al., 2024) and Mixtral-8x7B-Instruct (Jiang et al., 2024).

4.3 Experimental Setup

Data Splits. We evaluate all models on `ClaimFlow` using stratified train-validation-test splits constructed at the paper level to avoid leakage across citations. Papers are partitioned into 70-15-15 splits, and label distributions are preserved across splits.

Training Details. We fine-tune encoder-based models using hyperparameters selected on the validation set (details in Appendix B). LLMs are evaluated in zero-shot through four-shot prompting settings using a unified prompt template (details in Appendix D).

Setting	Model	P	R	F1
BASELINE	Majority	0.18	0.20	0.18
FINE-TUNED	BERT	0.51	0.50	0.50
	RoBERTa	0.73	0.71	0.70
	SciBERT	0.76	0.75	0.75
	DeBERTa	0.77	0.76	0.76
PROMPTED (4-SHOT)	GPT-3.5-turbo	0.75	0.75	0.75
	GPT-4o	0.78	0.78	0.78
	GPT-4.1	**0.82**	**0.81**	**0.81**
	LLaMA-3-70B	0.71	0.70	0.70
	Mixtral-8x7B	0.73	0.74	0.73

Table 3: Performance of different models on cross-paper claim-relation classification task.

Evaluation. Following Uma et al. (2021), for multi-class classification, we use macro-averaged precision, recall, and F1-score for label-based evaluation, aggregating performance across labels.

4.4 Results and Error Analysis

Main Results. Table 3 shows the results. Among fine-tuned encoders, RoBERTa consistently outperforms BERT, confirming the benefit of stronger pretrained representations. SciBERT further improves performance, suggesting that domain-specific pretraining on scientific text benefits modeling claim interactions. DeBERTa achieves the strongest encoder performance, indicating that improved contextual representation learning helps capture fine-grained claim relations. In contrast, the majority-class baseline performs substantially worse, showing that the task cannot be solved through label frequency alone.

In the zero-shot setting, all LLMs perform below most of the fine-tuned encoders, whereas few-shot prompting leads to consistent improvements across models. GPT-4.1 achieves the strongest overall performance among LLMs. Using the full input $(h_{\text{cited}}, s, h_{\text{citing}})$ consistently improves performance over variants that omit the citation context or one of the claims (ablation studies in Appendix G).

Error Analysis. Qualitative inspection of model errors in Table 12 (in Appendix G) reveals two dominant failure modes. First, models frequently confuse *extension* with *qualification*, particularly when a citing paper applies a prior claim to a new setting while simultaneously introducing constraints. Second, models struggle with cases where *support* or *refutation* is expressed indirectly. Hedging, cau-

Pipeline stage	P	R	F1
Claim identification	0.83	0.82	0.82
Relation classification	0.82	0.81	0.81
End-to-end graph recovery	0.81	0.79	0.80

Table 4: Stage-wise validation of the ClaimFlow-AutoGraph construction pipeline against expert annotations.

tious language, and implicit comparisons can obscure epistemic stance, leading models to default to the background label. Overall, the results indicate that while current models can capture coarse scientific stance, they struggle with nuanced claim-level reasoning.

4.5 ClaimFlow-AutoGraph

To enable large-scale analysis beyond expert annotation, we construct an automatically inferred claim graph over 13,358 papers from major ACL Anthology venues spanning 1979–2025. A three-stage pipeline first identifies candidate claim texts with a DeBERTa model fine-tuned on ClaimFlow, following Pramanick et al. (2025), then predicts a relation for each citing–cited claim pair with GPT-4.1 in a few-shot setting, and finally applies a canonicalization step that merges near-duplicate claim surface realizations (Figure 12; details in Appendices D.3, E.2, and E.3).

We note that the resulting claim graph serves as a scalable approximation that enables aggregate analyses of claim propagation and interaction patterns across NLP literature (Teodorescu and Mohammad, 2023).

4.5.1 Validation of ClaimFlow-AutoGraph

We validate each stage of the pipeline against the expert-annotated ClaimFlow test set.

Table 4 reports performance at each stage. *Claim identification* is evaluated as sentence-level binary classification; *relation classification* is evaluated on gold claim pairs and citation contexts, isolating the classifier; and *end-to-end graph recovery* evaluates the full pipeline, counting a predicted edge as correct only if the citing claim, cited claim, and relation label all match the gold annotation under the canonicalization procedure.

The end-to-end score of 0.80 F1 loses only 0.01 F1 relative to the gold-claim setting, indicating that the upstream identification and canonicalization stages introduce little additional error beyond the relation classifier itself. Canonicalization is likewise conservative: 200 sampled clusters show 0.92 purity and a 0.08 false-merge rate while reducing claim nodes by 11% (Appendix E.3). We further verify that the aggregate trends reported in §5 remain stable when edges are randomly removed, relation labels are perturbed, and canonicalization clusters are corrupted at realistic noise levels (Appendix E). Taken together, these results suggest that claim interaction structure can be recovered automatically at a scale far beyond what expert annotation supports, and that the longitudinal patterns we report are not artifacts of pipeline error.

5 Analyzing Claim Flow in NLP

Using ClaimFlow-AutoGraph, we analyze the evolution of NLP research along three dimensions: claim engagement by later work (§5.1), claim propagation across the literature (§5.2), and claim evolution over time (§5.3).

5.1 Claim Engagement

Takeaway. *NLP papers usually reuse prior claims as background or support. Challenges are rare and concentrated early in a claim's life cycle; these patterns are consistent across major venues.*

Q1. How do NLP papers engage with claims from prior work?

We analyze the overall distribution of relation types across ClaimFlow-AutoGraph, which provides a coarse-grained view of how claims are typically used when cited by later work.

Results. The majority of claim-claim interactions fall into the background (58.2%) and support (20.6%) categories, indicating that claims are most often reused as contextual premises or explicitly supported by subsequent work. extend relations (14.4%) occur less frequently, while qualify (4.5%) and refute (2.4%) are comparatively rare. This distribution closely follows the expert-annotated distribution in Table 2; Figure 15 in Appendix G.1 gives the full comparison.

Discussion. Results are consistent with long-standing norms in the NLP literature, where progress has typically been framed as cumulative rather than adversarial. Early symbolic and rule-based approaches (Church, 1988) were rarely rejected outright; instead, they were later positioned as foundational context when probabilistic and corpus-driven methods gained prominence (Brown et al., 1993). A similar dynamic accompanied the

shift from feature-engineered statistical models to neural architectures, where prior methods were generally contextualized or subsumed rather than explicitly refuted (Collobert et al., 2011). Even during major paradigm transitions, such as the move from phrase-based statistical machine translation to neural machine translation, earlier approaches were more often bounded or implicitly superseded than directly contradicted (Bahdanau et al., 2015).

Q2. How often are claims challenged in NLP research, and how are challenged claims subsequently engaged?

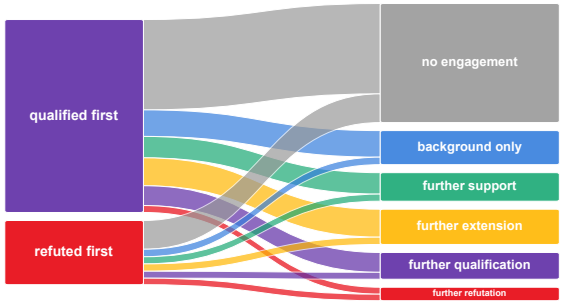


Figure 2: Distribution of post-challenge engagement for claims; flow widths indicate the number of claims.

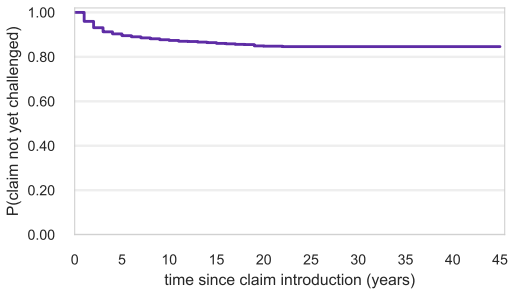


Figure 3: Survival curve for time to first claim challenge; claims not challenged during the observation window are right-censored.

We define *challenge* as interactions labeled `qualify` or `refute`, and analyze when claims are first challenged and how they are subsequently engaged. For each claim h, we measure the time to first challenge, defined as the difference between the publication year of the claim's originating paper and the earliest subsequent paper that challenges it. To examine what happens after the challenge, we identify all post-challenge interactions, i.e., claim–claim interactions occurring after the first challenge.

Results. We find that only a small number of claims are ever challenged (11.1%). Challenges take the form of qualification (8.3% of all claims) far more often than direct refutation (2.8%), indicating that claims are more often constrained or refined than rejected. Figure 2 shows that after an initial challenge, many claims receive no further substantive engagement (46.3%) or are cited only as background (13.2%). Among claims that continue to be engaged, further support (11.1%) and extension (13.8%) are more common than repeated qualification (10.4%) or refutation (5.2%), suggesting that challenges rarely lead to prolonged adversarial discourse. Figure 3 shows that challenges usually occur early in a claim's life cycle; among claims that are challenged, the median time to first challenge is 2 years.

Discussion. Results show that NLP research engages with challenged claims in a limited and non-escalatory manner. The survival curve reflects the tendency toward early vetting followed by normalization. Once a claim is qualified or refuted, later work tends either to move on or to re-incorporate the claim in a modified form, rather than continuing to contest it. For example, early debates around parsing models led not to repeated rejection of specific claims (Magerman, 1995; Collins, 1997), but to shifts in evaluation regimes and modeling constraints as statistical parsing matured (Charniak, 2000).

Two patterns further support this interpretation. First, uncertainty is concentrated early in a claim's life cycle: qualify and refute interactions are predominant within the first few years after a claim appears and decline as the claim ages, leaving long-lived claims with comparatively low cumulative uncertainty (Appendix A.2). Second, these patterns are consistent across publication venues: relation distributions are nearly identical across ACL, EMNLP, NAACL, EACL, AACL, and Findings, indicating that the observed trends cannot be attributed to any single venue (Appendix A.3).

5.2 Claim Propagation

Takeaway. *Most claims remain locally used, but the few that propagate tend to be taken up quickly.*

Q3. Do claims propagate across NLP papers or remain isolated?

For each claim h, we compute its propagation count, defined as the number of unique papers

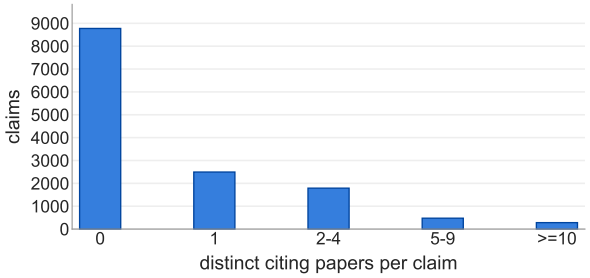


Figure 4: Distribution of claims by number of distinct citing papers.

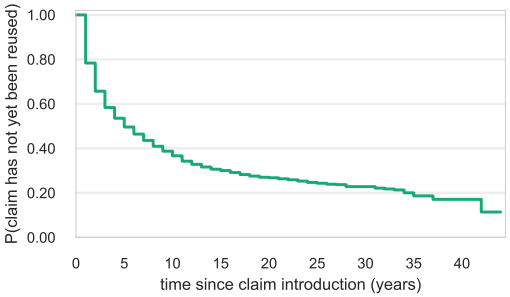


Figure 5: Survival curve for time to first claim reuse.

that reference h. Claims with a propagation count of zero are considered *isolated*, whereas claims referenced by one or multiple papers are considered *propagated*. Additionally, to analyze temporal reuse, we measure the time between a claim's introduction and its first subsequent reuse, treating unreused claims as *right-censored*.

Results. In Figure 4 we observe that most claims exhibit limited propagation across NLP papers, with 63.5% of claims remaining *isolated*. Among claims that propagate, most are reused by only a small number of subsequent papers (mean propagation count 1.37), while a minority propagate widely (∼2% reused by ten or more papers). Figure 5 further shows that claim reuse is time-concentrated. The probability that a claim has not yet been reused drops sharply within the first few years after its introduction.

Discussion. These engagement patterns show that influential ideas in NLP have often spread through rapid early uptake followed by consolidation, rather than gradual or sustained reuse. The early decline observed in the claim survival curve reflects this historical tendency toward early selection: claims that align with emerging datasets, tools, or modeling paradigms are quickly reused, whereas others effectively exit the active discourse without explicit refutation. For example, representational shifts – such as the introduction of word embeddings and later contextualized representations – were driven by a small number of claims that achieved early community adoption and subsequently dominated reuse patterns (Mikolov et al., 2013; Peters et al., 2018), while alternative proposals introduced in the same period attracted little long-term engagement (Collobert and Weston, 2008; Turian et al., 2010).

Q4. Do earlier claims shape later NLP research disproportionately?

For each claim h we define its temporal position and its exposure-normalized influence as

$$\begin{aligned}\text{AgeRank}(h) &= \frac{|\{h' \in \mathcal{H} : y(h') < y(h)\}|}{|\mathcal{H}|},\\ \text{NormInfluence}(h) &= \frac{\#\{\text{papers citing } h\}}{\#\{\text{papers after } y(h)\}}\end{aligned}$$

where $y(h)$ is the publication year of the paper introducing h, and lower `AgeRank` values correspond to earlier claims. `NormInfluence` measures *claim influence* as the number of distinct subsequent papers that engage with a claim, divided by its potential exposure time; this controls for the fact that earlier claims have simply had more opportunities to be cited than later ones. We analyze the association between the two quantities using rank correlations.

Results. We observe a negative association between temporal position and normalized downstream influence. Specifically, the Spearman rank correlation between AgeRank and NormInfluence is $\rho = -0.495$, indicating that earlier claims receive higher normalized engagement from subsequent papers compared to later claims. Because `NormInfluence` already divides by exposure time, this advantage cannot be explained by earlier claims simply having been available for longer.

Discussion. Results imply that claims introduced during periods of infrastructural or evaluative formation of NLP tend to exert long-lasting influence. For example, early claims associated with automatic evaluation metrics – most notably BLEU – rapidly shaped how machine translation progress was assessed and continued to structure downstream research long after their introduction (Papineni et al., 2002).

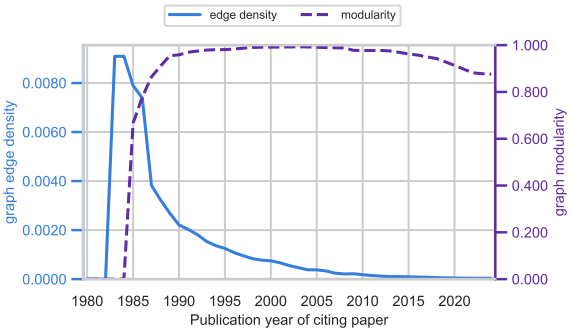


Figure 6: Structural evolution of the claim–claim interaction graph in NLP research.

5.3 Claim Evolution

Takeaway. *The claim graph stays sparse overall, yet increasingly organizes into thematic clusters that later partially reconnect through shared modeling paradigms.*

Q5. Does the claim graph densify or stratify over time in NLP research?

We construct a cumulative yearly claim–claim graph and track two structural properties: edge density, measuring how broadly claims connect, and modularity, measuring how strongly claims cluster into sub-communities. Formal definitions, separate density and modularity plots, and additional interpretation are provided in Appendix G.2.

Results and Discussion. Figure 6 shows that the edge density of the claim–claim graph remains extremely low overall (on the order of 10^{-3}), after an initial rise in the early years attributable to small-graph effects. Despite continued growth in the number of claims and relations, density stabilizes and slightly declines over time, indicating that new claims engage with only a limited subset of prior claims. At the same time, modularity increases steadily before stabilizing and declining in recent years. This suggests that NLP claims are increasingly organized around specialized task and method clusters, but have partly reconnected under shared modeling paradigms such as pretrained and foundation models.

A complementary analysis of the claim-level structural roles underlying these aggregate graph trends is provided in Appendix A.1.

5.4 Case Studies of Scientific Claim Evolution

To complement the aggregate analysis, we examine two influential claim trajectories. **BERT** (Devlin et al., 2019). Following its introduction, the original claim – deep bidirectional pretraining substantially improves downstream NLP performance – rapidly accumulated support and extension interactions as later work adapted pretrained representations to new tasks and domains. Over time, interactions increasingly qualified the claim through concerns about reasoning ability, efficiency, and computational cost (Kassner and Schütze, 2020; Ribeiro et al., 2020) (Figure 14(a), Appendix F). Rather than being overturned, the claim was progressively refined as the field evolved. **BLEU** (Papineni et al., 2002). Initially, the original claim – that BLEU provides an effective automatic proxy for machine translation quality – received strong support and widespread reuse as a standard evaluation metric in machine translation. Over time, later work increasingly qualified or refuted the claim by identifying limitations in semantic adequacy and correlation with human judgment (Novikova et al., 2017; Sellam et al., 2020) (Figure 14(b), Appendix F). Despite sustained criticism, the metric remained deeply embedded in NLP evaluation practice. We provide additional qualitative studies in Appendix F.

6 Conclusion

We introduced `ClaimFlow`, the first expert-annotated dataset for tracing relations between claims across papers, together with the task of *Claim Relation Classification* and an automatically inferred *claim graph* spanning decades of NLP research. Our analyses show that most claims remain isolated, explicit challenges are rare (and short-lived), early claims exert disproportionate influence, and influential claims are more often reshaped through qualification and extension than simply confirmed or rejected. Together, these point to a process of selective consolidation, in which a small set of claims becomes embedded in later research while many others leave little visible trace. We caution that this pattern should not be read as evidence that particular benchmarks, framing choices, or paradigms causally determine the direction of the field; our analyses do not test those mechanisms directly. By making claim-level interactions explicit, `ClaimFlow` provides a foundation for studying scientific knowledge as an evolving system of interconnected claims. We expand on these findings in Appendix H, and discuss applications of `ClaimFlow` together with directions for future work in Appendix I.

Limitations

Corpus scope. This study focuses on NLP papers drawn from the ACL Anthology, specifically venues under the ACL Events umbrella. While these venues represent a central and influential subset of NLP research, important contributions also appear in AI conferences, regional venues, and preprint platforms. As a result, the dataset does not capture the full breadth of scientific claims in NLP, and extending claim-level analysis beyond the ACL Anthology remains an important direction for future work.

Temporal coverage. Our longitudinal analyses aggregate over claims and relations within this curated corpus. Claims may continue to evolve beyond the temporal and venue boundaries of the dataset, and some influential ideas may be underrepresented due to sampling scope. Consequently, the trends we report should be interpreted as indicative patterns within a high-quality sample rather than an exhaustive account of the field.

Annotation scope. In addition, our annotations focus on abstracts, introductions, and conclusions, where authors typically articulate their central claims most explicitly. While claims may also appear in the main body of papers, annotating full texts would require substantially greater effort. Incorporating claims from the full paper content remains a future extension of this work.

Automatic labeling. Finally, our large-scale analyses rely on models trained on the human-annotated dataset to automatically identify claim relations in a broader corpus. It is important to acknowledge that no model achieves perfect accuracy, which can impact the quality of such analyses. However, Teodorescu and Mohammad (2023) showed that trend-level conclusions derived from large automatically labeled datasets tend to be highly accurate and show a strong correlation with trends identified through gold-label analysis, supporting the reliability and accuracy of our analyses.

Ethics Statement

Data and privacy. In this work, we use publicly available research papers from the ACL Anthology and do not involve personal data or human subjects. All annotations were performed on scholarly text and reflect interpretations of scientific claims rather than judgments about individual authors or research communities.

Responsible interpretation. We acknowledge that identifying and relating scientific claims involves interpretive decisions, and that scientific progress cannot be reduced to quantitative signals alone. The analyses enabled by `ClaimFlow` are intended to support reflective examination of research trends, not to prescribe scientific value or replace expert judgment. Any conclusions drawn from this work should therefore be considered in conjunction with qualitative, ethical, and social considerations that shape scientific practice.

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A Additional Analysis

A.1 Claim-Level Structural Roles

Q1. Do scientific claims primarily attract downstream convergence or integrate multiple prior claims?

We analyze the structural roles of claims in the directed claim-claim interaction graph, where nodes correspond to claims and directed edges represent

interactions from citing to cited claims. For each claim h, we compute its *in-degree* $k^{\text{in}}(h)$ as the number of distinct downstream claims that reference h, and its *out-degree* $k^{\text{out}}(h)$ as the number of distinct claims that h references.

We characterize *downstream convergence points* as claims with high in-degree and relatively low out-degree, indicating that multiple later claims converge on the same prior claim. Conversely, claims with high out-degree and relatively low in-degree *integrate multiple antecedents*: they draw on several prior claims while attracting fewer downstream claim links (Borgatti and Everett, 2000; Guimera and Nunes Amaral, 2005). To summarize this balance, we compute a signed degree-balance score for each claim $D(h) = \frac{k^{\text{out}}(h) - k^{\text{in}}(h)}{k^{\text{out}}(h) + k^{\text{in}}(h) + \epsilon}$, where negative values indicate convergence-dominated claims and positive values indicate integration-dominated claims. We set $\epsilon = 1$ to avoid division by zero and to assign neutral scores to isolated claims with no incoming or outgoing interactions. Because this smoothing also pulls low-degree claims toward zero, we use $D(h)$ as a descriptive balance measure rather than a calibrated centrality score.

Results. Figure 7 shows the distribution of $D(h)$. While substantial mass lies near zero, indicating weakly connected or comparatively balanced claims, the distribution is skewed toward positive values. Thus, claims more often integrate multiple prior claims than attract many downstream claim links. The negative tail nevertheless shows that downstream convergence points also exist, though they are fewer in number.

Discussion. This asymmetry suggests that NLP research exhibits a division of epistemic labor between integration and consolidation. Many claims synthesize and recombine existing ideas to explore new modeling choices, datasets, or problem formulations. Such integration is common during periods of rapid expansion, when researchers draw on multiple prior strands without immediately producing new focal claims – for example, during the early statistical turn, when diverse probabilistic assumptions and feature combinations were explored in parallel (Manning and Schutze, 1999).

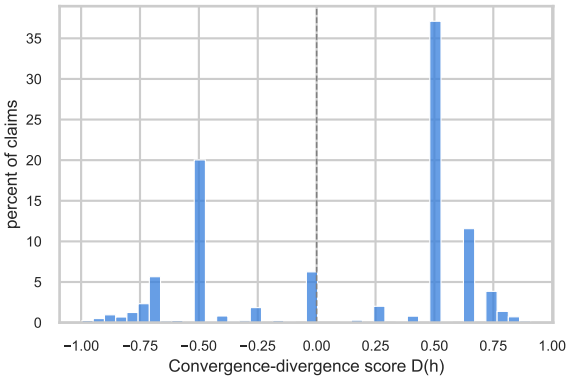


Figure 7: Distribution of signed degree-balance scores. Positive values correspond to integration of multiple antecedents.

At the same time, the smaller but substantial mass of convergence-dominated claims reflects moments of field-wide consolidation, where a limited set of claims becomes shared reference points for subsequent work. Claims associated with major resources or evaluation standards – such as the Penn Treebank (Marcus et al., 1993) or widely adopted automatic evaluation metrics (Papineni et al., 2002) – illustrate this pattern, attracting sustained downstream engagement while themselves referencing relatively few predecessors. More recently, pretrained language models similarly functioned as convergence hubs, rapidly drawing engagement from across tasks and subfields (Devlin et al., 2019).

Together, these findings indicate that NLP knowledge growth is neither uniformly integrative nor purely convergent. Instead, many claims combine several antecedents, punctuated by convergence onto a smaller number of stabilizing ideas.

A.2 Uncertainty Over Claim Lifetimes

Q2. Do scientific claims in NLP research accumulate uncertainty over time?

To examine whether claims accumulate uncertainty over time, we analyze the temporal trajectories of claim-claim interactions for each cited claim. We

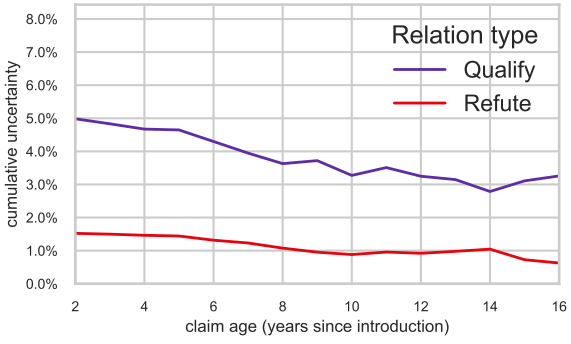


Figure 8: Cumulative uncertainty remains highest early in a claim's life cycle.

operationalize uncertainty using relations: qualify and refute, which respectively indicate that a claim's scope is constrained or that it is contradicted by later work.

For a claim h, we consider its sequence of downstream interactions ordered by publication year. We define the cumulative uncertainty at time t as the proportion of all interactions up to t that are labeled qualify or refute. We track how this proportion evolves as a function of claim age.

Results. Figure 8 shows that cumulative uncertainty remains low throughout a claim's life cycle, peaking shortly after introduction and declining thereafter. In the first few years after introduction, a non-trivial fraction of interactions involve qualify or refute, but as claims age, the proportion of uncertainty-inducing interactions declines, stabilizing at a low level for long-lived claims. Across all ages, uncertainty is dominated by qualify, while refute remains rare throughout.

Discussion. The observed decline is consistent with uncertainty in NLP claims being front-loaded rather than cumulative. New claims may be tested, bounded, or occasionally contradicted as the community probes their validity and scope; claims that remain active at later ages consequently exhibit a lower proportion of uncertainty-inducing interactions.

Historically, this pattern aligns with how influential ideas in NLP have matured. Early claims – such as those associated with new modeling paradigms or resources – often face scrutiny shortly after introduction, during which limitations are identified and scope conditions clarified. Once these boundaries are established, subsequent work typically builds on the refined claim rather than re-challenging it.

Taken together, these results indicate that uncertainty in NLP operates as an early-phase corrective mechanism rather than an ongoing process of erosion. Claims either stabilize after initial qualification or fade from active use, leaving long-lived claims with comparatively low cumulative uncertainty. This dynamic helps explain why explicit refutation remains rare and short-lived, and reinforces a broader picture of NLP research as progressing through early selection and path-dependent consolidation, rather than sustained adversarial debate.

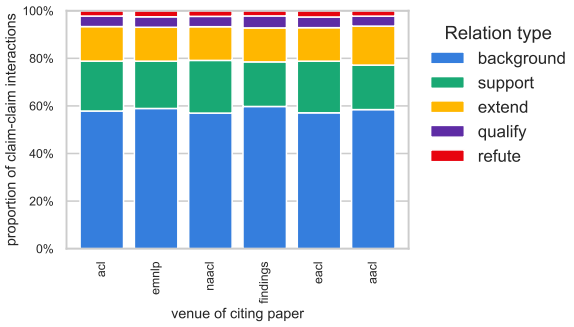


Figure 9: Distribution of claim–claim relation types across major NLP venues.

A.3 Venue-Level Claim Engagement

Q3. Do publication venues differ in how scientific claims are engaged in NLP research?

To examine venue-level differences in claim engagement, we group claim-claim interactions by the publication venue of the *citing paper*. For each venue v, we compute the distribution of relation types (support, extend, qualify, refute, background) across all interactions originating from that venue. To ensure comparability across venues with different publication volumes, we report normalized proportions (normalized by the number of publications) rather than raw counts.

Results. Figure 9 shows highly consistent engagement patterns across ACL, EMNLP, NAACL, EACL, AACL, and Findings. In all venues, background relations dominate ($\approx$ 57–60%), followed by support ($\approx$ 19–22%) and extension ($\approx$ 14–16%). Qualification remains uncommon ($\approx$ 4–5%), and refutation is rare ($\approx$ 2–3%). The small differences at the margins do not alter the overall ordering of relation types.

Discussion. The cross-venue similarity suggests that the aggregate engagement patterns are not driven by a single publication venue. Within the ACL Anthology venues studied here, venue choice modulates how claims are reused only at the margins and does not alter the dominant, cumulative mode of engagement.

B Additional Experimental Details

B.1 Training and Hyperparameters

During fine-tuning of encoder-based models, we perform grid search over epochs $e \in \{1, 2, 3, 4, 5\}$ and learning rates $lr \in \{1 \cdot 10^{-4}, 5 \cdot 10^{-5}, 1 \cdot 10^{-5}\}$

using a batch size of 32. Hyperparameters are selected based on validation performance.

For LLMs, we evaluate both zero-shot and four-shot prompting settings using a unified prompt template. Few-shot examples are sampled from the training split and held constant across models. Following common practice in in-context learning evaluation (Min et al., 2022), four-shot prompting is used as a lightweight demonstration setting rather than an exhaustive coverage of all relation types.

Each experiment is repeated three times. Across runs, performance variance remains below 0.02 macro-F1 for all models. We report Bootstrap confidence intervals over the test split in Appendix G.3.

C Annotation Instructions

This section provides detailed annotation guidelines used to construct `ClaimFlow`. The guidelines expand the conceptual framework described in §3 and specify the procedures used for claim identification, claim normalization, and claim-relation annotation.

C.1 Core Definitions

Annotators were first presented with the following definitions, which guided all subsequent annotation decisions.

Scientific Claim. A *scientific claim* is a central, empirically testable assertion about methods, data, or phenomena that a research paper advances as part of its contribution.
Claim Text. A *claim text* is a span of text in a research paper that explicitly expresses a scientific claim. The same underlying claim may be expressed by multiple claim texts within a paper.

Annotators were instructed to distinguish between *claims* as abstract scientific commitments and *claim texts* as their textual realizations.

C.2 Claim Identification

Annotators were provided with the *abstract*, *introduction*, and *conclusion* of each paper and instructed to identify claim texts from these sections. For each candidate sentence, annotators answered the following question:
Does this text express a scientific claim according to the definition above?
A text was considered a valid claim text if it:

- makes a testable assertion (rather than a definition or background statement),
- concerns a method, dataset, or empirical phenomenon,
- and reflects a central contribution emphasized by the authors.

When necessary, annotators were allowed to make *minimal edits* (e.g., resolving anaphora or removing references such as "this model") to ensure that claim texts were self-contained and interpretable in isolation. Annotators also recorded the section in which each claim text appeared. The citing claim text is not required to appear within the citation context itself and could occur elsewhere in the citing paper.

Examples of Valid and Invalid Claims. Examples of valid claim texts include statements such as:

- "BERT improves performance on multiple NLP benchmarks."
- "Scaling model size improves few-shot reasoning performance."

Examples that were not annotated as claims include:

- purely descriptive background statements,
- definitions without empirical assertions,
- or methodological details not presented as central contributions.

C.3 Claim Normalization

Multiple textual realizations within a paper may express the same underlying scientific claim. Annotators were instructed to group such realizations into a single canonical claim representation whenever the statements were semantically equivalent. Minor lexical, syntactic, or discourse-level variations were not treated as distinct claims.

C.4 Claim-Relation Annotation

In the second phase, annotators labeled relations between claims across papers.

Annotators were shown

- a *cited claim text* from a previously annotated paper,
- a *citing claim text* from a later paper,

- and the *citation context* in the citing paper (the sentence containing the citation marker together with the immediately preceding and following sentences)

When multiple citation contexts referred to the same cited paper, annotators selected the context that most directly expressed the interaction between the citing and cited claims.

Annotators then answered the following question:

Does the citing claim text substantively engage with the cited claim?

Substantive engagement was defined as explicitly evaluating, building upon, qualifying, or challenging the content of the cited claim.

When engagement was present, annotators assigned *exactly one* of the following relation labels:

- `support`: The citing claim provides evidence or results that reinforce the cited claim.
- `extend`: The citing claim generalizes or applies the cited claim to new settings, data, or tasks.
- `qualify`: The citing claim restricts the scope of the cited claim or specifies conditions under which it holds.
- `refute`: The citing claim presents evidence or arguments that contradict the cited claim.
- `background`: The cited claim is referenced for context or comparison without substantive evaluation.

Annotators were instructed to base their decisions on the explicit content of the claim texts and citation context, not on external knowledge.

C.5 Adjudication and Edge Cases

Ambiguous cases – such as partial engagement, mixed signals, or unclear scope – were flagged during annotation. Such cases were resolved through discussion, following the annotation guidelines and task definition described above.

Annotators were instructed to prioritize:

- explicit textual evidence over implicit assumptions,
- the claims under consideration rather than the overall contributions of a paper,
- and conservative labeling when evidence was insufficient to justify a stronger relation.

Relation	Cohen's κ
SUPPORT	0.80
EXTEND	0.73
QUALIFY	0.72
REFUTE	0.78
BACKGROUND	0.82
Overall agreement	**0.77**

Table 5: Relation-wise inter-annotator agreement.

C.6 Notes on Scope

Annotators were informed that the goal of the dataset is to make claim-level scientific interaction observable. As a result, the annotation focuses on clear claims and clear instances of engagement, rather than marginal or purely rhetorical references.

C.7 Inter-Annotator Agreement

Table 5 reports agreement for each relation type, measured with Cohen's κ on the shared subset of 1,001 doubly annotated papers. Agreement is highest for `background` (0.82) and `support` (0.80), where the citation context usually carries an explicit signal, and lowest for `extend` (0.73) and `qualify` (0.72). The two hardest relations for annotators are the same pair that models most often confuse (§4), suggesting that the boundary between applying a claim to a new setting and constraining its scope is genuinely difficult rather than an artifact of the label set.

D Prompting Large Language Models

D.1 Task Framing

Each instance consists of a cited claim text, a citing claim text, and the citation context from the citing paper. Models are instructed to select exactly one relation label from the predefined set: `support`, `extend`, `qualify`, `refute`, or `background`. Models are explicitly instructed to output only the relation label, without explanations or additional text.

D.2 Base Prompt Template

All LLMs are prompted using the same base template to ensure consistency across models. The prompt is shown in Figure 10.

D.3 Prompt for Automatic Graph Construction

For constructing `ClaimFlow-AutoGraph`, we use GPT-4.1 in a few-shot setting to classify each candidate citing–cited claim pair under a local citation context. The prompt follows the same

Instruction: You are given two scientific claims from NLP research papers and a citation context from the citing paper.

Cited claim: *{cited_claim}*

Citing claim: *{citing_claim}*

Citation context: *{citation_context}*

Task: Determine how the citing claim engages with the cited claim.

Relation labels:

- `support`: the citing claim provides evidence that reinforces the cited claim.
- `extend`: the citing claim generalizes or applies the cited claim to new settings or data.
- `qualify`: the citing claim restricts the scope or conditions under which the cited claim holds.
- `refute`: the citing claim contradicts the cited claim using opposing evidence or arguments.
- `background`: the cited claim is mentioned for context without evaluation.

Answer: Choose exactly one label from the list above. Output only the label.

Figure 10: Base prompt used for LLM-based claim relation classification.

label definitions as Figure 10, but makes the graph-construction decision explicit: if the citing claim does not significantly engage with the cited claim in the given context, the model must choose `irrelevant`. The template is shown in Figure 11.

D.4 Decoding and Post-processing

Models are decoded using deterministic or low-temperature settings to reduce output variability. If a model produces output that does not exactly match one of the valid labels, the output is mapped to the closest valid label when unambiguous; otherwise, the instance is marked as incorrect. No external tools, retrieval mechanisms, or chain-of-thought prompting are used.

E Additional Validation of `ClaimFlow-AutoGraph`

We provide additional analyses assessing the reliability of `ClaimFlow-AutoGraph` in end-to-end inference settings, including evaluations of automatic claim identification, lightweight canonicalization, and robustness to graph perturbations.

Instruction: You are constructing a claim-level graph of NLP research. Given a cited claim, a citing claim, and the citation context, first decide whether the citing claim significantly engages with the cited claim.

Cited claim: *{cited_claim}*

Citing claim: *{citing_claim}*

Citation context: *{citation_context}*

Decision rule: Use the citation context as local evidence for the interaction. If the citing claim does not significantly engage with the cited claim in this context, output `irrelevant`; do not force a `background`, `support`, `extend`, `qualify`, or `refute` label. If significant engagement is present, choose the relation that best describes how the citing claim engages with the cited claim.

Relation labels:

- `support`: the citing claim provides evidence that reinforces the cited claim.
- `extend`: the citing claim generalizes or applies the cited claim to new settings or data.
- `qualify`: the citing claim restricts the scope or conditions under which the cited claim holds.
- `refute`: the citing claim contradicts the cited claim using opposing evidence or arguments.
- `background`: the cited claim is mentioned for context without evaluation.

Few-shot examples: *{four_training_examples}*

Answer: Output only the label.

Figure 11: Prompt template used for relation prediction when constructing `ClaimFlow-AutoGraph`.

E.1 Pipeline Stages

Figure 12 summarizes the pipeline used to construct `ClaimFlow-AutoGraph`, outlined in §4. We describe each stage as follows.

Claim identification. First, we perform automatic claim identification at the sentence level to extract candidate claim texts from unannotated papers. Following Pramanick et al. (2025), we fine-tune DeBERTa on `ClaimFlow` and apply the resulting model to the larger ACL Anthology corpus, treating each detected sentence as an individual claim text.

Relation prediction. Next, for each citation between papers, we consider pairs of citing and cited claim texts and use GPT-4.1 in a few-shot setting to determine whether the citing claim engages with the cited claim and, if so, to predict the corresponding epistemic relation. When a paper cites another

Metric	Value
QUALITY Cluster purity ↑	0.92
RISK False merge rate ↓	0.08
SIZE Average cluster size	2.3
COMPRESSION Reduction in claim nodes	11%

Table 6: Evaluation of the lightweight canonicalization procedure.

paper multiple times, each citation mention yields a separate citation context, defined as the citation sentence together with its immediately preceding and following sentences, and claim interactions are predicted independently for each context (details in Appendix D.3).

Canonicalization. This process produces an automatically inferred directed claim graph in which nodes correspond to claim texts and edges represent predicted claim-claim relations. To reduce redundancy arising from repeated or near-duplicate claim texts, we apply a canonicalization procedure that clusters highly similar claim surface realizations and retains a single representative per cluster (Appendix E.3).

E.2 Automatic Claim Identification

We report the performance of the automatic claim identification model used as an upstream component in constructing the large-scale claim graph.

Experimental Setup. We fine-tune both SciBERT and DeBERTa on the expert-annotated `ClaimFlow` using the same train-validation-test splits described in §4. Claim identification is formulated as a sentence-level binary classification task predicting whether a sentence expresses a scientific claim. Performance is evaluated using macro-precision, recall, and F1-score.

Results. DeBERTa achieves the strongest overall performance (0.83/0.82/0.82 precision/recall/F1; Table 4), outperforming SciBERT (0.80/0.79/0.79) across all evaluation metrics. The strong performance of both models suggests that scientific claims can be identified reliably enough to support downstream construction of `ClaimFlow-AutoGraph`.

E.3 Lightweight Canonicalization

Because automatically identified claim texts may contain repeated or near-duplicate surface realizations within the same paper, we apply a lightweight canonicalization procedure to reduce redundancy in the automatically inferred graph.

Experimental Setup. Each claim text is represented using a sentence-level embedding obtained from a DeBERTa encoder fine-tuned for claim identification. Within each paper, we compute pairwise cosine similarity between claim embeddings and greedily cluster claim texts whose similarity exceeds a threshold of $\tau = 0.90$. For each cluster, we retain a single representative claim text corresponding to the longest surface realization, while edges incident on merged nodes are redirected to the retained representative. This procedure is intentionally conservative and merges only highly similar surface realizations.

Diagnostic Analysis. To evaluate whether canonicalization introduces substantial semantic distortion, we manually inspect 200 randomly sampled canonicalization clusters and compare graph statistics before and after clustering. Table 6 summarizes the results. We observe that canonicalization primarily merges near-duplicate claim surface forms while preserving the overall structural properties of the graph.

E.4 End-to-End Graph Recovery

Table 4 in §4 reports stage-wise validation of the pipeline. The end-to-end setting is the strictest of the three: a predicted edge counts as correct only if the citing claim, the cited claim, and the relation label all match the gold annotation after canonicalization, so a single upstream error in claim identification or a spurious cluster merge invalidates the edge even when the predicted relation is right. That this setting loses only 0.01 F1 relative to relation classification on gold claim pairs indicates that identification and canonicalization errors rarely compound into edge-level errors, which is the property the aggregate analyses in §5 depend on.

E.5 Robustness Under Perturbation

We additionally evaluate whether downstream analyses derived from `ClaimFlow-AutoGraph` remain stable under realistic levels of automatic inference noise.

Experimental Setup. We simulate perturbations by randomly removing edges, randomly altering relation labels, and perturbing canonicalization clusters at varying noise levels. For each perturbed

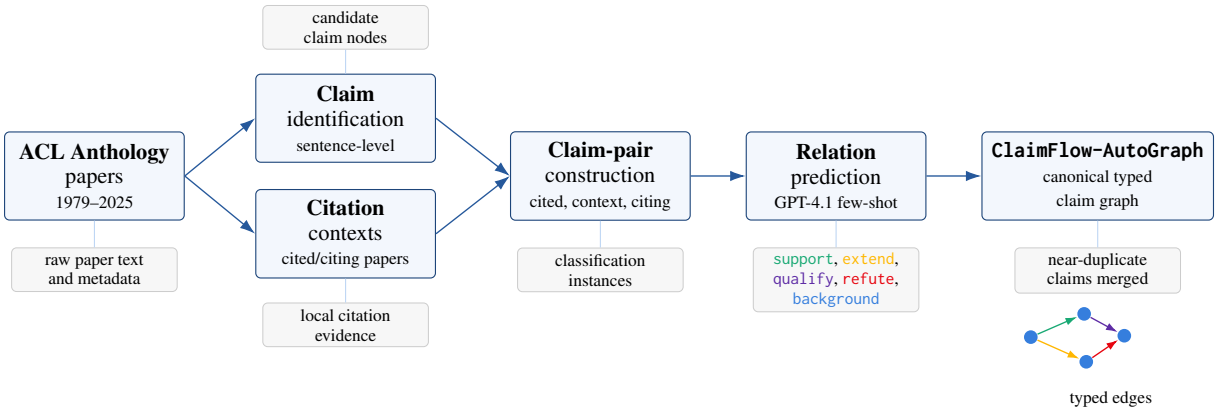


Figure 12: Pipeline for constructing ClaimFlow-AutoGraph.

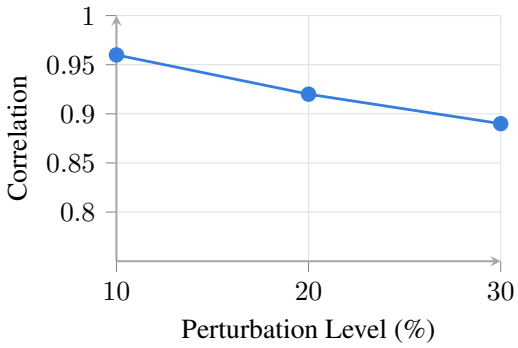


Figure 13: Correlation between original and perturbed aggregate graph statistics under increasing perturbation levels.

graph, we recompute the principal analyses presented in §5.

Results. Figure 13 shows the correlation between original and perturbed graph statistics under increasing perturbation levels. We observe that the principal qualitative findings remain stable under moderate perturbations, suggesting that the aggregate conclusions derived from ClaimFlow-AutoGraph are robust to realistic pipeline noise.

F Extended Case Studies of Scientific Claim Evolution

We provide additional qualitative analyses of representative claim trajectories extracted from ClaimFlow-AutoGraph. These case studies complement the aggregate analyses presented in §5 by illustrating how influential claims evolve through support, extension, qualification, and refutation over time. Representative downstream examples are shown in Tables 7 and 8.

F.1 BERT: Evolution of Bidirectional Pretraining Claims

We first analyze the evolution of the claim introduced by Devlin et al. (2019) that deep bidirectional pretraining substantially improves downstream NLP performance. Figure 14(a) visualizes a representative subset of downstream claim interactions associated with this trajectory, while Table 7 presents concrete examples extracted from ClaimFlow-AutoGraph.

In the years immediately following its introduction, the claim received a large number of support interactions from papers reporting strong empirical gains across diverse NLP tasks. Early downstream work further extends the original claim into new domains and modalities, including biomedical NLP, multilingual transfer, and multi-modal representation learning (Lu et al., 2019; Kanakarajan et al., 2019; Conneau et al., 2020). These interactions suggest that the claim rapidly became a convergence point for subsequent research, attracting engagement from a broad range of sub-fields within NLP (Liu et al., 2019; Clark et al., 2020).

As the trajectory evolves, later work increasingly introduces qualification interactions that constrain the scope of the original claim. Several downstream claims question whether pretrained bidirectional representations alone are sufficient for reasoning-intensive tasks or efficient adaptation under limited resources (Ribeiro et al., 2020; Kassner and Schütze, 2020). Other work shifts attention toward prompting-based and zero-shot paradigms (Brown et al., 2020; Sanh et al., 2022), partially challenging the assumption that task-specific fine-tuning of pretrained representations is the dominant route to generalization. For example, the zero-shot universal discriminator trajectory

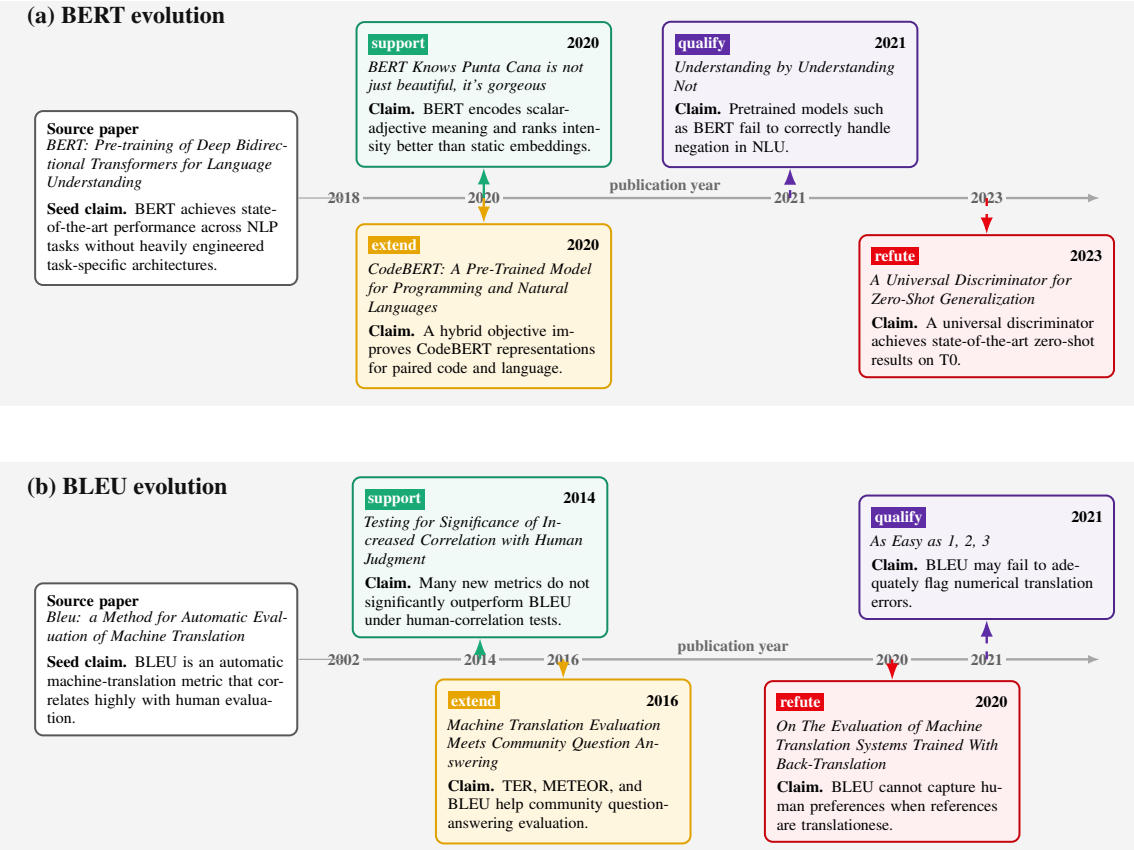


Figure 14: Timeline-centered claim evolution for two influential NLP papers. Each panel roots the original seed claim at the start of the publication timeline and places later interacting claims as branches at their publication years. The branch color and line style encode the ClaimFlow relation type: support, extend, qualify, or refute.

shown in Table 7 provides an explicit counterexample to BERT-style adaptation as the default mechanism for broad task transfer.

Importantly, the trajectory contains relatively little sustained adversarial rejection of the original claim itself. Instead, the dominant pattern is progressive refinement, where subsequent work extends, constrains, or contextualizes the original claim while preserving its broader influence on NLP research. This pattern closely aligns with the aggregate analyses in §5, which show that influential claims in NLP more often evolve through extension and qualification than through direct contradiction.

F.2 BLEU: Evolution of Evaluation Claims

We next examine the evolution of the claim introduced by Papineni et al. (2002) that BLEU provides an effective automatic proxy for machine translation quality. Figure 14(b) visualizes representative downstream interactions associated with this claim trajectory, while Table 8 provides concrete examples from `ClaimFlow-AutoGraph`.

Early interactions overwhelmingly support or reuse the original claim, reflecting the rapid adoption of BLEU as a standard evaluation metric in machine translation research (Koehn, 2004; Callison-Burch et al., 2007). During this period, BLEU functions not merely as a metric but as a shared evaluation infrastructure, enabling reproducible comparison across systems and benchmarks (Post, 2018). The support example shown in Table 8 further illustrates the metrics' durability: even later evaluation metrics often fail to significantly outperform BLEU under rigorous statistical testing.

Over time, however, later work increasingly qualifies and occasionally refutes the original claim. Several downstream claims identify important blind spots in BLEU's ability to capture semantic adequacy, numerical correctness, and human preferences (Novikova et al., 2017; Sellam et al., 2020). The numerical translation example shown in Table 8 demonstrates how high lexical overlap can preserve BLEU scores even when the meaning of a translation changes substantially. Other work directly challenges the metrics' validity in realistic

Year	Relation	Downstream claim	Interpretation
Seed claim: BERT achieves strong performance across NLP tasks without heavily engineered task-specific architectures.			
2020	SUPPORT	**Scalar adjective semantics.** *"BERT encodes rich knowledge about the semantics of scalar adjectives and provides better quality intensity rankings than static embeddings and previous models."* BERT Knows Punta Cana is not just beautiful, it's gorgeous	Gives a concrete linguistic case where bidirectional contextual representations outperform non-contextual baselines, supporting the representational part of BERT's claim.
2020	EXTEND	**Code and natural language.** *"Training CodeBERT with a hybrid objective function improves its performance."* CodeBERT: A Pre-Trained Model for Programming and Natural Languages	Extends BERT-style masked pretraining from ordinary text into paired programming-language and natural-language representations.
2021	QUALIFY	**Negation in NLU.** *"Current pre-trained language models like BERT fail to correctly handle negation in language understanding tasks."* Understanding by Understanding Not: Modeling Negation in Language Models	Makes the limitation explicit: high benchmark performance can coexist with brittle handling of logically decisive words such as negation.
2023	REFUTE	**Zero-shot universal discrimination.** *"The universal discriminator (UD) achieves state-of-the-art zero-shot results on the T0 benchmark."* A Universal Discriminator for Zero-Shot Generalization	Gives an explicit counterexample to BERT-style adaptation as the default path to broad task generalization: a universal discriminator can win in a zero-shot benchmark setting.

Table 7: Representative downstream interactions for the BERT case study. The examples illustrate the life cycle of a high-impact model claim: broad support, transfer into new modalities and domains, and later qualifications about reasoning, structure, and zero-shot alternatives.

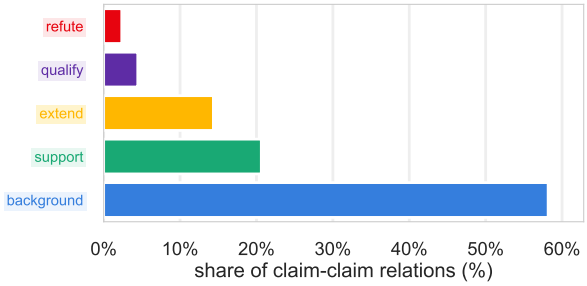


Figure 15: Distribution of claim–claim relations across `ClaimFlow-AutoGraph`.

translation settings by arguing that translationese artifacts can cause BLEU to reward outputs that humans do not prefer (Li et al., 2025).

At the same time, the trajectory also contains extension interactions that adapt BLEU-style overlap metrics to tasks outside machine translation, such as community question answering. This illustrates how influential methodological claims may continue to propagate into new application domains even as their original assumptions become increasingly constrained.

G Supplementary Results

G.1 Relation Distribution in `ClaimFlow-AutoGraph`

Figure 15 shows the full distribution of relation types across the automatically inferred graph, summarized in §5.1. The distribution closely tracks the expert-annotated distribution reported in Table 2: `background` accounts for 58.2% of interactions in both, and the remaining relations agree to within 1.1 percentage points (`support` 20.6% vs. 20.9%, `extend` 14.4% vs. 13.4%, `qualify` 4.5% vs. 5.4%, `refute` 2.4% vs. 2.1%). This agreement indicates that the automatic pipeline does not systematically over- or under-predict any relation type at scale, complementing the end-to-end validation in Appendix E.4.

G.2 Structural Evolution of the Claim Graph

For each year t, we construct a cumulative directed graph $G_t = (V_t, E_t)$, where V_t is the set of claims introduced up to year t, and E_t contains directed edges $(h_{\text{citing}} \rightarrow h_{\text{cited}})$, corresponding to claim-claim relations up to t. To assess whether the claim graph densifies over time, we compute edge density $\delta(G_t) = \frac{|E_t|}{|V_t|(|V_t|-1)}$, which measures the fraction of realized claim-claim interactions among all possible directed pairs. Intuitively, higher density

Year	Relation	Downstream claim	Interpretation
Seed Claim: BLEU provides an automatic MT metric that correlates highly with human evaluation.			
2014	SUPPORT	**Metric significance testing.** *"A high proportion of new metrics do not significantly outperform BLEU when evaluated with the introduced significance test."* Testing for Significance of Increased Correlation with Human Judgment	Supports BLEU's durability: even later metrics often fail to beat it significantly when correlation with human judgment is tested rigorously.
2016	EXTEND	**Community question answering.** *"Using machine translation evaluation measures such as TER, METEOR, and BLEU can enhance the performance of the cQA task."* Machine Translation Evaluation Meets Community Question Answering	Extends BLEU from MT evaluation into answer selection, where translation-style overlap becomes a feature for a different NLP problem.
2021	QUALIFY	**Numerical translation errors.** *"Existing de facto standard metrics such as BLEU may fail to adequately flag numerical translation errors."* As Easy as 1, 2, 3: Behavioral Testing of NMT Systems for Numerical Translation	Exposes a high-stakes blind spot: a single wrong number may preserve much of the n-gram overlap while changing the meaning completely.
2020	REFUTE	**Translationese references.** *"BLEU cannot capture human preferences because references are translationese when source sentences are natural text."* On The Evaluation of Machine Translation Systems Trained With Back-Translation	Directly refutes the proxy claim in a concrete MT setting: reference artifacts can make BLEU reward outputs that humans do not prefer.

Table 8: Representative downstream interactions for the BLEU case study. The examples show BLEU's transition from a durable automatic proxy to a cautionary baseline whose limits become visible in numeracy, generation, translationese, and human-preference settings.

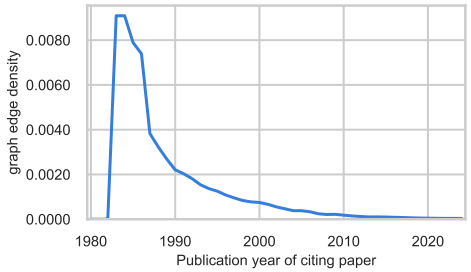
indicates that claims increasingly connect to many other claims, reflecting a more tightly interlinked body of work. To assess whether the graph stratifies, we measure modularity $Q(G_t)$, which quantifies the extent to which claims form groups with dense internal connections but relatively sparse connections to other groups. Intuitively, higher modularity indicates stronger clustering of claims into distinct thematic or methodological substructures.

Figure 16 separates the two trends summarized in the main paper. The density curve remains extremely low overall after the initial small-graph regime, indicating that new claims connect to only a limited subset of prior claims rather than forming a broadly interconnected graph. In contrast, modularity increases over time before plateauing and declining in recent years, suggesting that the claim graph becomes more stratified as the field specializes and later partially reconnects through shared modeling paradigms.

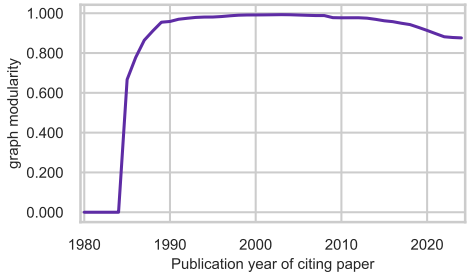
As the field expanded in the 1990s and early 2000s, the emergence of large resources and standardized evaluations, such as treebanks (Marcus et al., 1993), lexical databases (Miller, 1992), and task-specific benchmarks, encouraged specialization, leading claims to cluster around distinct tasks, representations, and evaluation regimes. The subsequent plateau and recent decline in modularity are consistent with methodological convergence in modern NLP. Since the mid-2010s, contextualized and pretrained models (Devlin et al., 2019) have been widely adopted across diverse tasks, increasing cross-subfield engagement and weakening previously isolated clusters. This trend has been further reinforced by multi-task and unified benchmarks (Wang et al., 2018, 2019), which explicitly promote cross-task comparison, and by the rise of foundation models (Bommasani et al., 2021), which frame many new claims as analyses or extensions of shared model families rather than task-specific innovations.

G.3 Label-wise Performance and Uncertainty

Table 11 reports label-wise `macro-F1` for the fine-tuned encoder and LLM configurations. To quantify how stable these per-label estimates are given the size of the test split, we compute approximate 95% confidence intervals by bootstrap resampling of the test instances. For GPT-4.1 (few-shot), the intervals are `background` $[0.83, 0.87]$, `support` $[0.79, 0.85]$, `extend` $[0.74, 0.82]$, `qualify` $[0.72, 0.82]$, and `refute` $[0.74, 0.86]$.



(a) Densification of the claim–claim graph over time.



(b) Increasing stratification of the claim–claim interaction graph over time.

Figure 16: Separate views of the density and modularity trends summarized in Figure 6.

Input	Model	Macro-F1
CONTEXT (s)	BERT	0.44
	RoBERTa	0.46
	SciBERT	0.51
	DeBERTa	**0.54**
CITED + CONTEXT (h_{cited}, s)	BERT	0.45
	RoBERTa	0.50
	SciBERT	0.56
	DeBERTa	**0.60**
FULL $(h_{cited}, s, h_{citing})$	BERT	0.50
	RoBERTa	0.70
	SciBERT	0.75
	DeBERTa	**0.76**

Table 9: Effect of different input components on claim relation classification performance.

Model	0-shot	1-shot	4-shot
GPT-3.5-Turbo	0.60	0.65	0.75
GPT-4o	0.61	0.70	0.78
GPT-4.1	**0.63**	**0.73**	**0.81**
LLaMA-3-70B	0.58	0.61	0.70
Mixtral-8x7B	0.60	0.65	0.73

Table 10: LLM performance (`macro-F1`) with different number of training examples.

H Extended Discussion

This section expands on the synthesis summarized in §5 and the Conclusion.

Claim-level propagation. Taken together, our findings suggest that scientific influence in NLP is unevenly distributed and only partially reflected by paper-level citation counts. Most claims remain isolated, whereas the relatively few claims that propagate tend to receive attention soon after publication. Earlier claims also receive greater normalized downstream engagement, even after accounting for their longer exposure to subsequent work. These results point to a process of selective consolidation in which a small set of claims becomes embedded in later research while many others leave little visible trace in the claim graph.

Relation dynamics and case studies. The relation-level analysis further clarifies how this consolidation occurs. Explicit challenges are rare and usually arise early in a claim's life cycle; when influential claims remain active, later work more often extends or qualifies them than directly confirms or refutes them. The BERT and BLEU case studies illustrate two forms of this process: a modeling claim can retain its broader influence while its scope is progressively refined, and an evaluation claim can remain infrastructurally important despite accumulating qualifications and occasional refutations. Scientific change in NLP therefore appears less as a sequence of clean replacements than as continued reuse under revised scope, evidence, and methodological context.

Field-level structure. At the field level, these local dynamics produce a claim graph that remains globally sparse while organizing into increasingly distinct clusters, followed by recent signs of reconnection around shared modeling paradigms. This pattern should not be interpreted as evidence that particular benchmarks, framing choices, or paradigms causally determine the direction of the field; our analyses do not directly test those mechanisms. Rather, it shows that claim-level interactions expose forms of uptake, refinement, and contestation that are obscured when citations or papers are treated as undifferentiated units. In this sense, `ClaimFlow` complements citation-based scientometrics with a representation of scientific influence as interaction among specific assertions.

I Applications and Future Work

Applications. The expert-annotated `ClaimFlow` provides a structured benchmark for evaluating whether NLP models can identify how scientific claims relate to one another. Claim Relation Classification is complementary to citation-intent classi-

Model	support	extend	qualify	refute	background
SciBERT	0.78	0.70	0.69	0.76	0.82
DeBERTa	0.80	0.73	0.71	0.78	0.83
GPT-4o (few-shot)	0.80	0.76	0.75	0.78	0.83
GPT-4.1 (few-shot)	**0.82**	**0.78**	**0.77**	**0.80**	**0.85**

Table 11: Label-wise macro-F1 scores for model performance on automatic claim-relation classification.

Error	Description	Typical confusion	Share
Extend–Qualify ambiguity	Difficulty distinguishing claim extension from scope restriction	extend ↔ qualify	∼30%
Implicit stance	Stance expressed through hedging or indirect comparison	support → background	∼20%
Evidence mismatch	Model aligns citing claim with the wrong cited claim	support → extend	∼15%
Weak refutation cues	Disagreement expressed without explicit negation	refute → qualify	∼15%
Context under-specification	Citation sentence lacks sufficient detail	any → background	∼10%
Other	Miscellaneous rare cases	–	∼10%

Table 12: Common error types observed in automatic claim relation classification.

fication and natural language inference: it focuses specifically on the epistemic stance that a later scientific claim takes toward an earlier one, grounded in its citation context. The results in §4 show that this form of reasoning is tractable, while persistent confusions between extension and qualification leave substantial room for better models.

The automatically inferred ClaimFlow-AutoGraph claim graph also creates opportunities for tools that operate over the scientific literature. Claim-centered navigation systems could help researchers follow how an assertion is reused, extended, bounded, or challenged across papers instead of returning only documents connected by citations. At an aggregate level, the graph could support further scientometric studies of idea diffusion, emerging research threads, and claims whose downstream engagement is unusually broad or contested. These are prospective applications rather than capabilities evaluated in this work, but the analyses presented here establish the claim-level structure needed to investigate them.

Future Work. An immediate direction is to expand ClaimFlow beyond ACL venues and NLP, allowing comparisons of claim evolution across publication communities and scientific domains. Broader coverage should include claims expressed throughout full papers, not only in abstracts, introductions, and conclusions, and should examine how corpus and venue boundaries affect the observed propagation patterns.

Improving automatic claim identification, relation prediction, and canonicalization would enable larger and more reliable claim graphs. Such graphs could be integrated into scholarly search and LLM-based research assistants, provided that predicted relations remain traceable to their source claims and citation contexts. Finally, combining claim-flow structure with evidence about model performance, robustness, datasets, and benchmarks could test whether early interaction patterns predict which claims persist or how research trajectories develop. We leave these predictive and causal questions to future work.